
Asserted Zeros: When a Valid Guarantee Certifies an Unmeasured Quantity

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Abstract

1 A distribution-free guarantee certifies whatever score it is handed, including one
2 that was never measured. We call this a *measurement-integrity failure*, and docu-
3 ment one in a pipeline whose two LLM judges grade spoken answers. It stored
4 option probabilities to five decimals, so anything below 5×10^{-6} became exactly
5 0.0, and rescaling what survived to sum to one pushed the one remaining option
6 (usually the correct one) to 1.0. A record’s score is one minus the true answer’s
7 stored probability, so on 90.6% of graded records that score was exactly 0.0: a
8 certainty no judge expressed. Those scores became the held-out set that fixes the
9 release threshold, as conformal prediction does. You sort them, take the one sitting
10 at a position your error budget picks, and release an option whose score falls at or
11 below it. With nine in ten at the manufactured 0.0, that position lands among them,
12 so the threshold is 0.0 and the gate admits exactly the options a judge rounded up
13 to 1.0. It now turns on whether the two judges rounded up to the same option,
14 not on the evidence beneath them. A pre-registered probe found every grading
15 decision we checked unchanged. On the recordings this system actually ran, the
16 gate released nothing at all, with or without the defect. Re-imposed at that resolu-
17 tion on correctly measured scores, the rounding makes the gate release more than
18 measurement does in 36 cases of 42 and fewer in none, certifying answers hon-
19 est measurement would have withheld. The repair is dull: measure every option
20 or refuse. Five questions about how probabilities are recorded catch this before
21 calibration.

22 1 Introduction

23 Verification systems compose formal guarantees over quantities a model produces. The guarantee
24 is proved about a random variable; the system computes something it takes to be that variable; and
25 the composition is sound only if the two are the same. Verification checks the guarantee against the
26 score it was handed; nothing validates that the score is the quantity intended. We show that they
27 can differ: a guarantee can remain mathematically valid while certifying a quantity that was never
28 observed. We call this a *measurement-integrity failure*: the downstream guarantee remains valid, but
29 the quantity supplied to it is not the quantity the system intended to measure. The category is broad:
30 a verifier fed a truncated context measures every probability correctly and still supplies the wrong
31 conditional, so we say which member we demonstrate and which our diagnostic catches: a pipeline
32 *recording a value it never observed*, because its record format has no way to represent *unmeasured*.

33 The setting is the one this workshop asks about: agentic systems delegate consequential decisions to
34 LLM verifiers, and a natural way to make that verification *reliable* is conformal prediction: calibrate
35 a nonconformity score (ours is $1 - p$ for an option, so a small score means a confident grader) on
36 held-out examples, release only when the calibrated uncertainty set is decisive [Vovk et al., 2005,

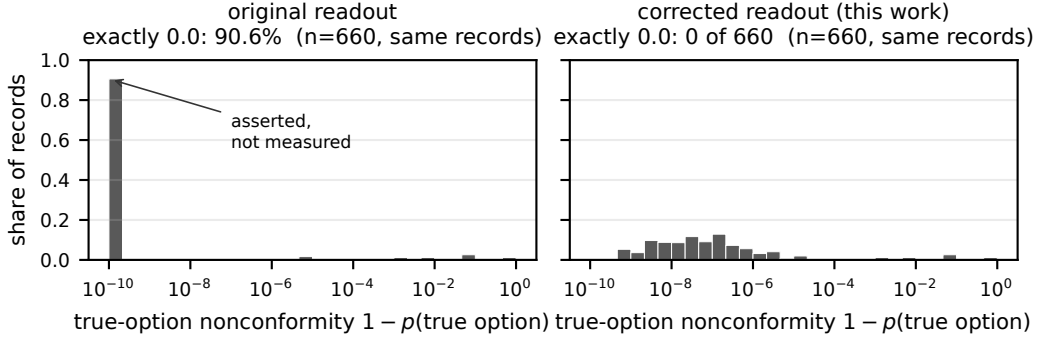


Figure 1: True-option nonconformity on the same 660 graded records, before and after. Left: the original readout, whose spike at exactly 0.0 is asserted, not measured. Right: the same records re-read by the corrected readout of §3, same models, same prompt; 0 of 660 scores are exactly zero (minimum 4.1×10^{-10}). Axes are shared, and exact zeros are drawn at the 10^{-10} axis floor, since a log axis cannot show zero.

Angelopoulos and Bates, 2023]. The guarantee is distribution-free and finite-sample, attractive precisely where verifier stakes are highest. Our case study is a speech-based assessment verifier; the contribution is the failure mode, not the domain.

That guarantee attaches to *whatever score is fixed in advance*. Conformal validity says nothing about whether the score measures what its user thinks it measures. Consider one record from our corpus. The verifier’s recorded distribution over $\{a, b, c\}$ was $\{1.0, 0.0, 0.0\}$, giving the true option a score of $1 - 1.0 = 0.0$: total certainty. Measured properly, the same model on the same input gives $\{0.999998, 1.35 \times 10^{-7}, 2.24 \times 10^{-6}\}$, a score of 2.37×10^{-6} . Neither zero was observed, and neither was the 1.0: 0.999998 rounded to five decimals *is* 1.0. One such record is a rounding curiosity. Ninety percent of them is a calibration set whose defining feature, where its mass sits, is manufactured (Figure 1).

This failure needs only three properties, and none of them is exotic: a *lossy record of a small probability* (a fixed-decimal writer that rounds it away, a token matcher blind to the spelling it arrived under, a top- k window that omits it, or any combination), *renormalization over whatever is left*, and a downstream consumer that treats the result as a probability. None looks like a defect on its own; each is a reasonable local choice, and our own first repair of the writer re-introduced the same defect at six decimals (§3). Reading the source of three widely used option-scoring interfaces and one API specification (Appendix G): bounded visibility is documented, and restricted-set normalization appears in the harnesses, legitimately there, since the candidate set is enumerated by the task rather than lost, which is exactly what our case lacked. We claim nothing about prevalence.

Conformal prediction is chosen precisely for its indifference to the model’s probabilities: it needs no assumption that scores are calibrated or even meaningful, only exchangeable, and that immunity is what makes this failure quiet: practitioners who adopt conformal methods *because* confidences cannot be trusted have little reason to audit them afterwards. The immunity covers calibration, not measurement.

The system. Two instruments appear below: the artifact of §2 was found on an 11-item work-readiness screener, and the corrected readout and cost analysis run on a 10-item civics corpus built after it. Four words carry the counting: an *item* is one question of the instrument in play; a *record* is one channel rendering of one spoken answer to one item; a *session* is one speaker’s full set of answers; and the *verdict* is the session-level pass/fail the decision rule returns. Short written answers to civics questions (the Powergrading corpus; Basu et al., 2013) are spoken by synthetic voices [rany2], degraded through four channel conditions, and transcribed with faster-whisper large-v3 [SYSTRAN, Radford et al., 2023]. Two LLM judges, seated as *judge* (Qwen3-32B-AWQ; Qwen Team, 2025) and *evaluator* (Gemma-3-27B-IT; Gemma Team, 2025) and served by vLLM 0.25.1 [Kwon et al., 2023], then grade each transcript independently, each with one decoding call whose first token is read as the decision. Option probabilities are read from that token’s top- k log-probabilities and com-

73 bined across judges by minimum nonconformity; a split-conformal threshold per item turns scores
 74 into per-item uncertainty sets; and a deterministic validator (the *gate*) *releases* a verdict for a ses-
 75 sion, rather than withholding it, only if every combination reachable from those sets yields the same
 76 verdict: equivalently, if the smallest and largest attainable totals fall on the same side of the mark.
 77 That rule is a counting rule: pass if at least 6 of the 10 items are correct; for a k -item subset the mark
 78 scales with k , so a single item is its own decision. Per-item budgets compose by the union bound
 79 ($\sum \alpha = k \cdot \alpha_i$ at a common α_i), and the exchangeability unit is the speaker; Appendix A gives the
 80 arithmetic.

81 The readout is lossy in three places: a fixed-resolution record, a token matcher that recognizes only
 82 some spellings, and a shared window, which §2 takes apart. The window is shared with the whole
 83 vocabulary: the constraint parameter our harness passed does not exist in the serving stack’s version
 84 and is silently ignored, so the corpus was extracted unconstrained.¹ The object of study is not the
 85 pipeline but the *measurement* that enters calibration.

86 **What conformal calibration does, and why an atom breaks it.** Split conformal holds out ex-
 87 amples whose answers are known, scores each one, sorts those scores, and reads a threshold off the
 88 sorted list, the $\lceil (n+1)(1-\alpha) \rceil$ -th, where α is the error budget its user chooses. At test time an option
 89 joins the released set if its score falls under that threshold, and a verdict is released only when the
 90 set is decisive. The guarantee is real but narrow: coverage of at least $1 - \alpha$ for *whatever score was*
 91 *fixed in advance*, assuming nothing about whether that score is calibrated or even meaningful. That
 92 indifference is the method’s selling point, and exactly what makes this failure quiet: a threshold read
 93 off a sorted list is defenceless against a pile of identical values. When ninety percent of the scores
 94 are the same manufactured 0.0, the quantile lands inside that pile and the gate stops responding to
 95 evidence, while every guarantee it advertises still holds.

96 Two literatures ask whether a model’s confidence can be *trusted*; the conformal literature offers guar-
 97 antees that hold regardless. We ask a third question: whether the number recorded as a confidence
 98 was *observed at all*, and show that when it was not, the guarantee is untouched, the judge’s deci-
 99 sions are unchanged, and the defect surfaces only in the calibration layer. What we add is not the
 100 concept, which measurement theory and missing-data statistics have long carried, but its instance:
 101 this failure inside a distribution-free guarantee, whose own indifference to the score keeps it hidden.
 102 The remedy is correspondingly boring: measure every option or refuse.

103 Concretely, we contribute:

- 104 1. **An exact-zero artifact in a verification signal** (§2): on 52.3% of judge and 80.1% of
 105 evaluator extractions only one option is left with a nonzero record, so its probability be-
 106 comes 1.0; under min-combination across two judges, 90.6% of graded records’ true-option
 107 scores are exactly 0.0, and the calibrated threshold is exactly 0.0 from $\alpha = 0.096$ upward.
 108 Re-running the original readout on a retained window separates the causes: repairing the
 109 five-decimal writer alone removes 7,733 of them, the token matcher alone 256, the top- k
 110 window alone *none*; counts that overlap rather than partition (Table 1).
- 111 2. **Correct measurement, independently checked** (§3): a readout that measures every op-
 112 tion’s probability or refuses (absence raises; full-precision serialization), validated on
 113 13,992 records (0 exact zeros) and, for both extractors, against a full-vocabulary recon-
 114 struction that shares neither its transport nor its window.
- 115 3. **A pre-registered probe** (§4) separating what the artifact distorted (calibration inputs) from
 116 what it did not (argmax decisions), including the one substantive falsification and five vac-
 117 uous ones at operating points where nothing releases; all ten are reported (Appendix C).
- 118 4. **What it costs** (§5): re-imposing the defect on measured scores drives the calibrated thresh-
 119 old to an endpoint on every item it reaches (141/141 item-splits). Above the step (more
 120 than ten of the n_{cal} calibration grades wrong) the gate releases nothing at all. Below it, on
 121 the split reported, it issues certificates at a budget of 0.10 that measurement does not. At
 122 this floor we claim the endpoint, not a direction of release; at the writer’s own 5×10^{-6}
 123 release rises in 36 of the 42 item-splits reached and falls in none. At $\tau = 10^{-2}$ aggregate
 124 error *falls* while this happens (19 of 20 resplits), so watching it would read the defect as an
 125 improvement; at the writer’s own 5×10^{-6} it *rises* in 13 of 20 (Appendix F).

¹No recorded number is affected: masking shifts surviving log-probabilities by a constant, so the renormalized distribution is invariant, and the leading token is an option on 346 of 347 audited records regardless.

126 5. **A diagnostic another team can apply** (§7): five questions about a readout that expose
 127 the failure we demonstrate before its output reaches a calibration set, including the one our
 128 own defect got past.

129 We frame this for the workshop’s question at the level where we found the failure: the *verification*
 130 *of the verifier’s evidence*. The system is a judge/evaluator pipeline inside an agentic assessment
 131 loop, and the failure sits in its verification layer: it went on returning well-formed verdicts while the
 132 evidence beneath them was fabricated, with nothing in its behaviour to separate the two.

133 2 The artifact: asserted zeros

134 The two corpora here predate the civics corpus: an 11-item spoken *work-readiness* screener (three
 135 options per item) rendered as degraded synthetic speech: the same 77 utterances under two channel-
 136 and-voice draws, 770 and 1,848 extraction rows, of which 660 and 1,584 are graded bank records
 137 (Appendix E). The original readout took option probabilities from the single decision token’s top-20
 138 log-probabilities, renormalized over the options present, and wrote the result to disk rounded to five
 139 decimals. Three routes in that pipeline reach the same record. An option can be absent from the
 140 window; present but spelled in a way the reader’s token matching does not recognize; or counted
 141 correctly and still fall below 5×10^{-6} , which the serializer writes as 0.0. What unites them is not
 142 the window: *a value that could not be represented was written as the value zero*.

143 Which route acted is measurable, because the readout is re-runnable: we re-extracted both corpora
 144 keeping the raw window, cut it to the top 20 by rank, aggregated under the old token matcher,
 145 renormalized, rounded to five decimals, and that reconstruction reproduces the recorded pattern of
 146 zeros on every record that has one. Two questions come apart, and only the second is causal: what
 147 did the reader have in hand when each zero was written, and what would repairing one factor alone
 148 change?

Table 1: Which factor erased each zero first, and which zeros repairing that factor alone would remove. The columns count the same 7,861 zeros and must not be added: the left one partitions them, the right one cannot, since a zero may be removable by several repairs or by none alone.

	Factor	Zeros it erased first	Zeros that repairing it alone removes
149	Writer, five decimals	7,733	7,733
	Token matcher, bare letters only	97	256
	Window, top-20 rank cut	31	0

150 Every zero that some single repair removes is removed by repairing the writer; the other 128 need
 151 the writer *and* a second factor. Repairing the window removes not one of the 7,861 zeros: wherever
 152 an option fell outside it, the writer would have erased it anyway. The rank window is not merely
 153 secondary but entirely preempted. The token matcher is a real contributor: 97 zeros where the
 154 option sat in the window under a spelling the reader could not match, and 256 that repairing it alone
 155 removes. This artifact is a record format that could not represent a small probability, primarily by
 156 rounding it away, in a minority of cases by never counting it.

157 The consequence follows mechanically whichever route acted. When only one option is left with a
 158 nonzero record, its probability becomes 1.0, by renormalization where the others were never counted,
 159 by the writer where they were counted and rounded away. The record’s resolution is the limit of what
 160 it can distinguish: across all 7,861 recorded zeros we re-scored, the largest true probability behind
 161 one is 1.84×10^{-5} and the median is 2.55×10^{-8} . Nothing above 10^{-4} was ever hidden, but every-
 162 thing below the writer’s precision was, and that is exactly the range a calibration threshold reads. On
 163 the two corpora that is 52.3%/51.4% (judge) and 80.1%/79.5% (evaluator) of all extraction rows,
 164 the two figures in each pair being the two corpora (770 and 1,848 rows).

165 That is the per-judge picture. The pipeline then combines two separately-prompted judges by mini-
 166 mum nonconformity ($s(o) = 1 - p(o)$; min across judges), so either judge collapsing to a one-hot
 167 on the true option forces the combined true-option score to exactly 0.0. On the graded records this
 168 occurs on **90.6%/89.7%** of rows, the spike visible in Figure 1. The combiner amplifies: because
 169 either judge collapsing to a one-hot *on the true option* is enough to force the combined score to zero,

the combined share exceeds either judge’s own. The two judges are one model family per role, so their errors are correlated and we do not predict the combined rate from the marginals.

The calibration consequence is immediate: a split-conformal threshold is the $\lceil (n+1)(1-\alpha) \rceil$ -th order statistic of the calibration scores. Once the zero count reaches that order statistic the threshold is exactly 0.0: at a 90.6% zero share, $\alpha \geq 0.096$; at 89.7%, $\alpha \geq 0.104$. Crossing it does not change the released set, since the five-decimal grid leaves no score between 0 and 10^{-5} : an item’s set holds the options recorded at 1.0 and excludes the rest. A set that ends up empty is promoted to the full option set, withholding rather than asserting. At a threshold of 0.0 the set is therefore whichever options some judge scored 1.0: a singleton when the judges’ one-hots agree, a doubleton when they disagree on a three-option item, and the full set when neither survives. The gate appears sharp; the sharpness is the artifact.

We emphasize what this is not. It is not a violation of conformal validity: the coverage guarantee of §1 holds here. What the artifact destroys is its *sharpness*. Exact coverage additionally requires a continuous score distribution; ours carries an atom of 90.6% at exactly 0.0, so realized coverage is pinned near 0.906 for every α above the crossover: conservative, and nearly insensitive to the budget its user chose. Ties are a known pathology, and smoothed conformal randomizes through them by construction [Vovk et al., 2005]; it would restore a non-degenerate threshold here without restoring any evidence, since the atom it spreads was manufactured rather than observed. And 0.906 is not a calibration property: at threshold 0 the true option is covered exactly when some judge recorded it at 1.0, so coverage is the rate at which a grader’s survivor is the truth. The gate has become a detector of grader agreement. Nor is it usefully described as an implementation bug, though it began as one. Here the proxy was a rounded, renormalized window rather than a distribution, and everything downstream (thresholds, set sizes, release decisions) inherited that while remaining formally correct. That is the measurement-integrity failure of §1, in its narrow form: a value recorded that was never observed. The defect is invisible to conformal machinery because it is upstream of it, and invisible to ordinary testing because, as §4 shows, it changes almost nothing a system’s behaviour would reveal.

3 Measuring instead of asserting

The corrected readout makes one decoding call per record and computes each option’s probability as the *sum over its token variants* (“a”, “A”, “(a)”, “a.”) in the decision token’s top- k ($k=100$), since a model spells one choice several ways and any single spelling understates it. Two hard guards define the semantics: (i) an option with no variant anywhere in the window raises an error: absence is never recorded as zero, which was the equation that produced the artifact of §2; (ii) the unnormalized option mass is persisted, so the residual carried by non-option tokens is visible rather than silently absorbed. The score is computed on the renormalized distribution; the unnormalized mass is recorded beside it and never enters the score.

The refusal rate is the natural cost of “measure or refuse”, so we report it: zero. Both caches hold exactly 13,992 rows and the probe holds no unscorable records, a check rather than an assertion, since the readout raises and nothing catches it, so one refusal would have halted the run mid-file.

Two incidents in our own deployment show how easily this defect is re-introduced, even by a team actively looking for it. First, the decoding constraint we believed we were imposing never was: our harness passed a parameter name the serving stack had retired, and unknown parameters are ignored at every arity rather than rejected, so the readout measured an unconstrained model throughout. A readout that scores each option in its own call (how we first rebuilt it) inherits that assumption and fabricates a plausible near-uniform distribution, silent exactly in the borderline band that decides calibration; an identity guard (the emitted token must equal the option asked for) turned $\sim 28,000$ would-be silently wrong scores into one diagnosed defect. Second, our own first writer rounded to six decimals, returning a measured 4.3×10^{-7} as exactly 0.0 on disk: *the same defect as §2, re-manufactured by the fix itself*. A smoke test caught it; a regression test pins it now. That is no coincidence: the writer is the dominant route into the original artifact too, and it returned the moment we stopped watching.

We then built a second, larger corpus *under the measured readout from the start*: 350 synthetic speakers $\times$ 10 short-answer items $\times$ 4 channel conditions (13,992 records after excluding two un-speakable responses, disclosed in the pre-registration), graded over a binary {correct, incorrect} option domain by the same two blind judges. This is a different corpus and option arity than §2’s

224 corpora, and the *same-record* before/after lives in Figure 1 and the probe of §4; re-extracting the
 225 original corpora under the corrected readout leaves 0 of 660 (and 0 of 1,584 on the second corpus)
 226 at exactly zero, from minima of 4.1×10^{-10} and 5.2×10^{-10} . It is the substrate for §5. Under the
 227 measured readout: exactly-zero true-option scores number **0/13,992**; no option is recorded at ex-
 228 actly 1.0; per-extractor minimum probabilities are 7.8×10^{-9} and 3.9×10^{-9} ; and the unnormalized
 229 option mass separates the judges: the first never records below 0.99 (minimum 0.99987), the second
 230 has a real tail (36 values below 0.9, minimum 0.4379), so where variants leave mass elsewhere, the
 231 readout now says so instead of absorbing it.

232 We can also report where the options sat in the window: on an audited sample of each extractor,
 233 each option’s *best-spelled* variant has median rank 2 or better, never exceeds 4, and none of the
 234 1,388 ranks falls outside a top-20 window (a statement about where an option is *findable*; most
 235 individual spellings sit far lower). For this extractor on this two-option corpus, then, a rank cut at 20
 236 could not have produced the artifact at all: the same negative result §2 reaches on the three-option
 237 corpora, where repairing the window removes none of the zeros.

238 **Checking the fix against something other than itself.** A readout that validates only against its
 239 own output is exactly what this paper warns about, so for *both* extractors we reconstructed the
 240 same quantity from the full vocabulary, outside any server, on 347 records each (Appendix D). Four
 241 predictions were registered before the measurement; all four held on both. Option mass agrees
 242 with the deployed readout to a median ratio of 1.000000 (99th-percentile deviation 0.013 evaluator,
 243 6.6×10^{-7} judge); scores agree to median ratios of 1.000000092 and 1.000000000, every one within
 244 a factor of two; no score is exactly zero; and the spellings the readout aggregates over carry all but
 245 6×10^{-12} (evaluator) and 2.3×10^{-7} (judge) of the option mass.

246 Four limits, stated rather than left to be found. The reconstruction runs on the same inference engine
 247 as the deployment, so it validates the *readout* and not the engine, and says nothing about serving
 248 stacks we did not use. Second, it shares the corrected readout’s token matcher, so it cannot detect
 249 a spelling that matcher misses; that class is bounded separately, by enumerating the vocabulary for
 250 option spellings the matcher rejects. Third, those missed spellings carry more mass on the judge
 251 than the evaluator (2.3×10^{-7} against 6×10^{-12} , above the smallest option probabilities we report),
 252 so we checked them against what they could perturb rather than against a threshold: on no record
 253 does that mass reach the smallest option’s own mass, at worst 13% of it. Fourth, on a deliberately
 254 adversarial stratum the evaluator’s agreement loosens by an order of magnitude (mass ratios 0.90 to
 255 1.16) while the judge’s does not; we report that stratum apart rather than pooling it into the headline.

256 4 A pre-registered probe: what the artifact did and did not distort

257 Ten predictions are registered across two documents (4 + 6), and all ten are reported: four held, one
 258 was substantively falsified, and five were falsified *vacuously*, at registered operating points where
 259 nothing releases at all (Appendix C). Both, one for the probe below and one for the corpus of §3,
 260 were frozen before the harness that scores them existed, an ordering visible in the artifact’s history
 261 and timestamped by our own record-keeping rather than a third party. The one that carries weight
 262 here is the corpus document’s fourth, registered before the corpus existed: that after the fix the share
 263 of calibration scores at exactly 0.0 would fall below 0.20 (from ~ 0.90) and the calibrated threshold
 264 would be strictly positive in every split seed. It held, its operating points amended before scoring
 265 (Appendix C).

266 Before re-extraction we froze four predictions about what correct measurement would change on
 267 the *original* corpora (200 records each, both judges, stratified by *recorded support*: the number of
 268 options recorded nonzero, so support-1 rows are exactly those collapsed to a one-hot). Three held;
 269 one failed; we report all four. They were not independent: Q1 is close to forced by the fix and Q4
 270 follows from it; Q2 and Q3 were the two genuinely at risk.

271 **Held.** Q1: on every support-1 row whose surviving option was the true one, the measured score is
 272 strictly positive (100/100, 156/156, 99/99, 162/162 across pool $\times$ judge): the zero was always an
 273 artifact. Q3: argmax decisions are robust: 0/200 flips on every stratum, which bounds the flip rate
 274 on the probed sample at 1.5% (exact one-sided 95%; the rule of three). We do not extrapolate that
 275 bound to the full corpus: the probe was stratified by recorded support and drawn from the original
 276 corpora, so it is not a random sample of anything larger. “Unchanged” is what we observed, not what
 277 the probe guarantees. Q4: the calibration dial comes alive: the $\alpha=0.1$ threshold moves from 0.0 to

278 1.3×10^{-3} (1.7×10^{-4} on the second corpus), both on the probe’s stratified 200-record sample, an
 279 illustration of the dial rather than an operating point, with zero-score share $\sim 0.90 \rightarrow 0.000$.

280 **Failed.** Q2 predicted the measured distributions would differ substantially in total variation (median
 281 $TV \geq 0.05$ on support-1 rows). Measured: median TV runs 4.2×10^{-8} to 1.1×10^{-4} across corpus,
 282 extractor and stratum: falsified by two to six orders of magnitude. This falsification is the sharpest
 283 result of the four: the artifact barely moved the distribution’s *bulk*; what it destroyed was the *tail*.
 284 The practical reading is uncomfortable for downstream users: a system built on the original readout
 285 makes the same decisions and reports the same argmaxes, so nothing in its behaviour signals that its
 286 calibration layer is consuming fabricated evidence. The artifact is invisible except where it acts.

287 Stated plainly: the original readout was accurate to about 10^{-6} nearly everywhere, and its whole
 288 defect was writing an exact 0 where the truth was around 10^{-6} . That sounds negligible, and for the
 289 model’s decisions it is, but not for calibration, for the reason §1 gives.

290 5 What the artifact costs at the gate

291 The probe says the artifact is hard to see. This section asks what it does, by re-imposing the readout’s
 292 own equation on the measured scores of the binary corpus of §3: record any option below a floor τ as
 293 0.0, renormalize, and hold the records, the split, the calibration and the decision rule fixed. This is a
 294 controlled simulation of the mechanism and is *not* pre-registered. The shape is right: a fixed-decimal
 295 writer zeroes every probability below its resolution, so the original defect *was* a probability floor, at
 296 5×10^{-6} , and this re-imposes the same operation, at that setting and coarser ones. It does not model
 297 either other route: the token matcher drops mass rather than rounding it, and the rank route removes
 298 no zeros when repaired (Table 1) and is inert here anyway. At the writer’s own 5×10^{-6} the floor
 299 already zeroes 53.1% of calibration scores and already issues wrong verdicts measurement does not
 300 ($57 \rightarrow 59$), while no item loses all its release. The figures below use $\tau = 10^{-2}$, a coarser setting
 301 that shows the mechanism’s shape; Appendix F runs the ladder between them.

302 **Where truncation acts, it drives the threshold to an endpoint.** A floor rewrites only the scores
 303 beneath it, so it misses some items: on four of the ten here (calibration error 0.060–0.069; median
 304 three across the resplits below) enough of the calibration sample already sits above τ that the quan-
 305 tile never moves, and threshold, release set and error count are bit-identical to the measured gate.
 306 Truncation is not uniform; it reaches items whose evidence was concentrated enough for a floor to
 307 reshape.

308 Where it does reach, most of the calibration sample collapses onto 0 or 1 and the $(1-\alpha_i)$ quantile
 309 lands in the mass the floor created, so the threshold takes one of two values and a single comparison
 310 decides which. The threshold is 0 exactly when at least $\lceil (n+1)(1-\alpha_i) \rceil = 106$ of the 116 calibration
 311 scores are zero: equivalently, in the collapsed regime where the floor has driven every reached score
 312 to 0 or 1, when both judges miss on at most *ten* of them. The count is the invariant, not the rate: the
 313 speaker-level split leaves $n_{\text{cal}} \in \{115, 116, 117\}$ and the order statistic allows exactly ten at each,
 314 so $10/116 = 0.086$ is this split’s instance of a threshold that moves slightly with n . Below that
 315 line, below the *step*, the gate admits every option the readout zeroed: release rises on all four items
 316 concerned, and it rises *including on sessions the system gets wrong*. Above it the quantile falls in
 317 the mass at one, every set is full, and the gate releases nothing at all. Figure 2 shows both regimes
 318 at $\Sigma\alpha = \alpha_i = 0.10$, with the collapse falling between the items at error 0.069 and those at 0.155,
 319 consistent with a step at 0.086, though ten items cannot separate 0.086 from α_i itself, and we claim
 320 only that the step lies where the order statistic requires.

321 **Certificates the measurement does not issue.** Each is a single-item decision, so a release is a
 322 verdict the system issues. Below the step, truncation issues wrong verdicts that measurement does
 323 not: items 1, 2 and 8 go from 0 to 2, 1 and 1, item 7 from 3 to 4. This is the paper’s title under a
 324 controlled floor: a guarantee that remains valid, certifying sessions on probabilities that were never
 325 observed. Above it, at this floor, the harm runs the other way: items 20 and 3 (pool IDs, not ranks)
 326 lose all 81 and all 34 of their releases, so a floor coarse enough to reach them makes the gate useless
 327 on hard items and reckless on easy ones.

328 **When aggregate monitoring would not catch it.** Summed over the ten decisions at $\tau = 10^{-2}$,
 329 wrong verdicts *fall* under truncation, 57 to 44, because the two high-error items stop releasing at all:
 330 a practitioner watching aggregate error would see improvement while per-item behaviour became

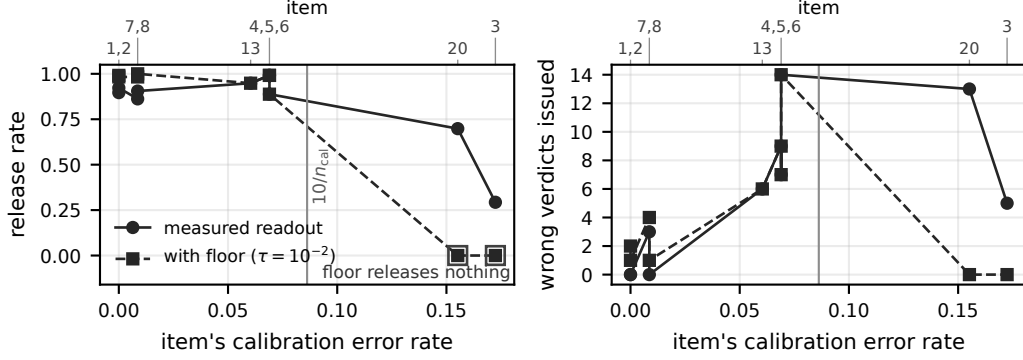


Figure 2: Ten single-item decisions at an honest budget ($\Sigma\alpha = \alpha_i = 0.10$, telephone stratum, $n_{\text{test}}=116$), with the floor re-imposed at $\tau = 10^{-2}$ (coarser than the five-decimal writer’s own 5×10^{-6} ; Appendix F runs the ladder), ordered by the item’s calibration error rate: the share of calibration records both judges score below 0.5 for the true option, a close proxy for the share the floor zeroes (which needs both below τ ; 1.57% of records fall between). The vertical line is the step, at ten wrong of n_{cal} . The floor reaches six of the ten: above the step it releases nothing (marked), and on the four below it releases more than measurement does, extra certificates that are not free (right panel). The other four keep enough calibration mass above τ that the quantile never moves, so the two readouts coincide exactly.

arbitrary. That inversion belongs to the coarse floor, not the defect: at the writer’s own 5×10^{-6} truncation makes aggregate error *worse* in 13 of 20 resplits, and the sign flips only once the floor silences the hard items and their wrong verdicts leave the total. The error rates driving the step are themselves channel-driven: the share of calibration records both judges score below 0.5 runs 5.4% to 42.9% across our four conditions, tracking word-error rate (1.15 on those records against 0.094 elsewhere), so the recording, not the grader alone, sets how much error a gate must tolerate.

What survives resampling, and what does not. These counts come from one split, at an n_{cal} where §6 puts the training-conditional band at $1.4\text{--}2.5\times$, so we repeated the measurement over twenty speaker-disjoint resplits (seeds 0–19), varying only the seed. They resample one corpus, so they bound split sensitivity rather than replicate. Its first third, reserved for fitting a grader, is unused here, leaving n_{cal} and n_{test} near 116. At $\tau = 10^{-2}$ three things hold: the threshold lands on an endpoint on every item the floor reaches (141/141 item-splits); at least one item collapses to zero release in 20/20; and in 20/20 at least one item below the step issues a wrong verdict that measurement does not: the claim this section is named for. At the writer’s own 5×10^{-6} , two of the three survive: the endpoint result is scale-invariant (27/27, 42/42, 58/58, 141/141 up the ladder), and in 13/20 splits (a different 13 from the aggregate count above) an item below the step still issues a wrong verdict measurement does not, release rising in 36 of the 42 item-splits the floor reaches and falling in none. The collapse does not: no item loses all release below 10^{-2} .

Two things do not. The constants move: measured wrong verdicts 38–72 across splits, truncated 26–63, so $57 \rightarrow 44$ is an instance, not an estimate. And truncation below the step does *not* always release more than measurement: of the 117 item-splits where the floor reaches an item whose error is at most ten in n_{cal} , release rises in 105, ties in 6, falls in 6. At this floor we therefore claim the endpoint, not a direction of release. The step’s location is likewise not a clean function of error rate: item 20 at median error 0.134 collapses in 19/20 splits, item 3 at 0.138 in 5/20. What decides it is where an item’s score mass sits relative to τ . The counts are single-*instance* quantities, so we re-ran the measurement on a fresh serving process: every per-item threshold is bit-identical, the inert and zero-release sets unchanged, one release count moving by one.

And at the deployed operating point, none of this is visible. There the rule is $k=10$ at $\Sigma\alpha \leq 0.5$, so $\alpha_i \leq 0.05$; at $\alpha_i=0.01$ the order statistic $\lceil 115.83 \rceil = 116 = n$ is the sample maximum for *any* score distribution, moving interior as α_i rises. On the mixed-condition stream the system actually ran, nothing releases with or without the artifact, at any τ and every registered budget. That invisibility is a property of the deployed *stream*, not of the budget alone: on the telephone stratum

at the same operating point, 32 of 116 sessions release under measurement and none under the floor: there, the artifact would have been a visible collapse.

6 Limitations and related work

Limitations. At calibration $n=116$ a single split’s realized rates are measurements, not tests of the guarantee. Miscoverage conditional on one calibration draw is Beta-distributed in n and the order statistic alone [Vovk, 2012], with a 95th percentile at $2.5\times$, $2.0\times$ and $1.4\times$ nominal for $\alpha = 0.01$, 0.02 , 0.10 : twice nominal is an ordinary draw here, a property of the calibration size, not of our scores, and one to quote as a band. Our corpus is synthetic-TTS speech over one assessment domain with one model family per seat, and the constants do not carry beyond it (Appendix B records how each is checked). The mechanism (a probability recorded through a representation that cannot hold it, renormalized, then consumed as a calibration score) is not specific to speech or assessment. Appendix G cuts the other way: what we found elsewhere was not what was causal here (repairing the window removes not one zero), and the mechanism behind most of our own zeros appeared in neither system that writes a probability record. The failure is reachable, not shown to be common.

Related work. Conformal prediction and its split form are classical [Vovk et al., 2005, Angelopoulos and Bates, 2023]; training-conditional validity is due to Vovk [2012], risk control to Angelopoulos et al. [2021], and class-conditional feasibility floors to Ding et al. [2023]. Conformal methods over LLM outputs typically consume log-probabilities as scores [Kumar et al., 2023, Quach et al., 2024], credal-set predictors inherit coverage the same way [Javanmardi et al., 2024], and set-valued classification traces to Sadinle et al. [2016].

A second literature asks whether model confidences can be trusted at all: networks are overconfident [Guo et al., 2017], language models carry signal about their own correctness [Kadavath et al., 2022], verbalized confidences can beat raw logits [Tian et al., 2023], and uncertainty over generated text needs semantic equivalence [Kuhn et al., 2023]. A third studies LLM verifiers themselves, now standard evaluation infrastructure [Zheng et al., 2023] with documented biases such as candidate order [Wang et al., 2024].

Older fields asked it first. Measurement theory separates the quantity intended from the one an instrument delivers; construct validity [Cronbach and Meehl, 1955] is the concern that a well-behaved instrument measures the wrong thing. Coding an absent value as zero is a missing-data mechanism, not at random here: an option is dropped exactly when it is small [Rubin, 1976]. The closest analogue is runtime verification, a trusted checker guarding an untrusted component [Sha, 2001].

7 A diagnostic for measurement integrity

Measurement-integrity failures are hard to test for because they leave behaviour intact: nothing in the stack raised an alarm. What can be tested is the readout, and five questions expose the one we found. *Is every option’s probability present in the window you read?* A readout returning fewer options than exist is asserting the rest. *Do you keep the unnormalized mass?* Renormalizing without recording what was renormalized destroys the evidence anything went missing. *Can you name a recorded probability you never observed?* A measurement pipeline should be able to refuse; ours raises rather than write a zero it never measured. *Can the format you write to represent the smallest probability you can measure?* A window admitting every option is undone by a writer that rounds what it admits: five decimals cannot express 10^{-7} , and repairing that writer alone removes 7,733 of our 7,861 zeros. *Does the emitted token match the one you asked for?* This catches a constraint passed but silently ignored, as ours was.

All five inspect the readout, which needs its code. The distribution it produces needs only the scores: an atom like Figure 1’s is visible at a glance, and is the check a consumer of someone else’s harness can still run. These target the subclass we demonstrate: a value recorded without being observed, not a readout that measures the wrong conditional correctly. Conformal prediction did not fail here. A valid guarantee is only as meaningful as the measurement handed to it, and nothing inside checks the handoff.

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510 A The decision rule, in full

511 The release rule is a counting rule: a session passes if at least 6 of its 10 items are correct. For
512 the k -item subsets used in §5 the mark scales as $\max(1, \text{round}(6k/10))$, giving 1, 1, 2, 3, 6 at $k =$
513 1, 2, 3, 5, 10. Per-item budgets compose by the union bound: a session is mis-set if any of its k item
514 sets is, so $\Sigma\alpha = \sum_i \alpha_i = k \cdot \alpha_i$ at a common α_i . This needs no independence between items,
515 which matters because ours are not independent. The exchangeability unit is the speaker: splits are
516 speaker-level, so no speaker’s records straddle calibration and test.

517 B The committed artifact

518 Every quantitative claim in this paper is recorded against the committed artifact it came from, and
519 a build-time check re-reads the compiled PDF and requires each value printed in the content pages
520 to trace to one of those records; the figures are produced by committed scripts that re-derive their
521 headline shares from the artifacts on each build. An anonymized code and data release accompanies
522 this submission.

523 All model serving runs on a single 46 GB GPU with one extractor resident at a time; the two ex-
524 traction passes over the binary corpus take under an hour in total. Every other result in this paper
525 is CPU-only arithmetic over the committed score caches and re-runs in minutes, so the analysis
526 reproduces without a GPU.

527 A second fabrication path exists in the same code: if *no* option survived the window, a one-hot was
528 fabricated on the emitted token. Its rate on the original run is unmeasurable, because that run did
529 not keep the window; re-extracting the same records with the window kept bounds the opportunity
530 instead, and across all 5,236 extractions there is none where the old reader would have found nothing
531 to renormalize. A real defect that never fired here.

532 C Pre-registration bookkeeping

533 Of the ten registered predictions, five, all loop-policy predictions in the corpus document, were
534 falsified *vacuously*: at their registered operating points nothing releases at all, for the reason §5
535 gives and not because of the loop, so we count them apart from the one substantive falsification (Q2
536 in §4): a prediction refuted by an inert operating point is information about the configuration, not the
537 hypothesis. The corpus document’s fourth prediction had its scored operating points amended before
538 scoring, from $\Sigma\alpha \in \{0.05, 0.10\}$ to $\{0.10, 0.20\}$, because at the lower budgets nothing releases and
539 the prediction would have been decided by arithmetic rather than evidence; the amendment is dated
540 in the registered document and is additions-only.

541 For completeness of the registration record, a third frozen document (a feasibility-frontier study, five
542 predictions) is registered in the artifact and is not reported in this paper. It ships with the artifact; its
543 predictions are scored there.

544 D How the full-vocabulary reconstruction works

545 For each sampled record we take the prompt the deployed extractor sent, assert its token ids are
546 identical to the server’s own tokenization of the same messages, and run one forward pass outside
547 any server. The next-token distribution is read at the decision position over a window of 2048 of
548 a 151,669- and 262,145-token vocabulary. Option mass is the sum over all tokens that normalize
549 to an option. Some option spellings fall outside that window on every record, so tokens outside it
550 are *bounded* rather than assumed zero: each contributes at most the window’s smallest probability,
551 giving a relative bound of 1.45×10^{-13} (evaluator) and 5.3×10^{-11} (judge). The comparator is
552 a fresh extraction from the same server instance rather than the committed cache, because a long-
553 running server does not reproduce its own earlier outputs exactly: 154 of 200 records bit-identical
554 against the day-old instance that wrote the cache, worst deviation 1.27×10^{-2} . A clean relaunch, by
555 contrast, reproduces the cache almost perfectly (99.98% and 99.62% of 13,992 rows bit-identical per
556 extractor), so the instability is within-lifetime state, not launch nondeterminism; the same-instance
557 comparator guards against both.

558 E The corpora of §2

559 Both corpora serve an 11-item spoken work-readiness screening instrument (options a/b/c per item;
 560 the scoring key is transcribed from a deployed system and pinned in the artifact). Utterances are
 561 rendered by synthetic voices [rany2], degraded, and transcribed with faster-whisper large-v3 [SYS-
 562 TRAN]. The first corpus: 77 utterances (66 answer-bank, 11 clean probes) $\times$ 5 channel levels (tele-
 563 phone band-limiting and additive noise at signal-to-noise ratios +5/0/ - 5/ - 10 dB) $\times$ 2 noise
 564 seeds = 770 rows, of which 660 are graded bank records. The second: the same 77 utterances $\times$ 8
 565 accented voices $\times$ 3 levels (telephone; SNR 0/ - 5) = 1,848 rows, 1,584 bank. The binary civics
 566 corpus of §3 uses 4 channel draws (telephone; SNR 0/ - 5/ - 10).

567 F The τ -ladder

568 §5’s figures use $\tau = 10^{-2}$. The same measurement at the defect’s own scale, on the same stream
 569 and split:

Table 2: §5’s per-item measurement repeated at four floors, including 5×10^{-6} , the writer’s own resolution.

τ	Cal. scores at 0.0	Gates changed	Zero-release	Wrong
0 (measured)	0%	—	0	57
10^{-6}	46.3%	1	0	57
5×10^{-6} (the writer’s)	53.1%	2	0	59
10^{-4}	75.8%	3	0	60
10^{-2} (§5)	91.3%	6	2	44

571 At the writer’s own resolution the floor already zeroes half the calibration mass and the gate already
 572 issues wrong verdicts that measurement does not (57 $\rightarrow$ 59); the zero-release collapse, and with it
 573 the fall in aggregate error that makes the defect read as an improvement, appear as the floor coarsens.
 574 Repeating §5’s twenty resplits at each rung gives the same separation: endpoint thresholds at 27/27,
 575 42/42, 58/58 and 141/141; no zero-release item below 10^{-2} ; and aggregate error worse under the
 576 floor in 0, 13, 15 and 1 of 20 splits as τ coarsens. Post-hoc, as §5 states.

577 G A lightweight audit of related interfaces

578 To ask whether the assumptions our failure needs are peculiar to us, we read the source of three
 579 widely used option-scoring interfaces and one API specification on 2026-08-19. All four are mov-
 580 ing targets and we did not record the commits we read, so the reading is dated rather than pinned: a
 581 reader cloning later may find any cell changed. The table records assumptions, not defects. Restrict-
 582 ing to an enumerated candidate set and normalizing over it is correct where the measured quantity is
 583 explicitly conditional; that is ordinary scorer behaviour and is not the failure this paper documents.
 584 We make no claim about prevalence, and none about the correctness of any system below.

Table 3: Four option-scoring interfaces, each read at the level its Evidence column states: [V] we read the implementation, [D] we read only documentation or a type declaration. A dash means the property was not found on the path we read.

System	Evidence	Precision	Candidate visibility	Renormal.	Interpretation
OpenAI chat API	spec text [D]	—	capped at 20	—	bounded visibility is documented in the specification at 20, the same width we read the restricted-set assumption exists, and is legitimate here: the choices are enumerated, so the value is conditional by construction
lm-eval-harness	source [V]	full float in the record	exact (echo); raises on the chat path	over the task's choices	likewise legitimate: the candidate set is defined by the task, not lost; no precision loss on the serialized record
lighteval	source [V]	full float in the record	—	over the task's choices	the visibility limit exists in configuration; its multiple-choice path parses text and never consumes probabilities
inspect_ai	config [D], path [V]	—	0–20 declared	none	

Two results are worth separating. Bounded visibility and restricted-set normalization each appear outside our system, so the assumptions are not ours alone, though wherever we found the normalization, it was over a candidate set the task enumerates, which is precisely what our case lacked. But the route that produced most of our own zeros, a writer that cannot represent a small probability, appears in *neither* harness that writes a probability record: `lm-eval-harness` and `lighteval` serialize in full float, and `inspect_ai`'s multiple-choice path writes none. And in one case the discipline §3 recommends is already practised: on its chat-completions path `lm-eval-harness` raises rather than returning an approximation, giving the unavailability of prompt log-probabilities as its reason. We report that as behaviour only; the source says nothing about calibration, so we claim no shared motivation.

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Justification: §6 states the training-conditional inflation ($1.4\text{--}2.5\times$ nominal α at $n=116$), the synthetic-TTS single-domain scope, and that the artifact mechanism is expected to transfer while the constants are not.

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